Generalized Function

A generalized function, also called a "distribution" or "ideal function," is the class of all regular sequences of particularly well-behaved functions equivalent to a given regular sequence. As its name implies, a generalized function is a generalization of the concept of a function. For example, in physics, a baseball being hit by a bat encounters a force from the bat, as a function of time. Since the transfer of momentum from the bat is modeled as taking place at an instant, the force is not actually a function. Instead, it is a multiple of the delta function. The set of distributions contains functions (locally integrable) and Radon measures. Note that the term "distribution" is closely related to statistical distributions.

Generalized functions are defined as continuous linear functionals over a space of infinitely differentiable functions such that all continuous functions have derivatives which are themselves generalized functions. The most commonly encountered generalized function is the delta function. Vladimirov (1971) contains a nice treatment of distributions from a physicist's point of view, while the multivolume work by Gel'fand and Shilov (1964abcde) is a classic and rigorous treatment of the field. A result of Schwarz shows that distributions can't be consistently defined over the complex numbers $\mathbb{C}$.

While it is possible to add distributions, it is not possible to multiply distributions when they have coinciding singular support. Despite this, it is possible to take the derivative of a distribution, to get another distribution. Consequently, they may satisfy a linear partial differential equation, in which case the distribution is called a weak solution. For example, given any locally integrable function f it makes sense to ask for solutions u of Poisson's equation

$$\nabla^2 u = f \tag{1}$$

by only requiring the equation to hold in the sense of distributions, that is, both sides are the same distribution. The definitions of the derivatives of a distribution $p(x)$ are given by

$$\int_{-\infty}^{\infty} p'(x) f(x)\,dx = -\int_{-\infty}^{\infty} p(x) f'(x)\,dx \tag{2}$$

$$\int_{-\infty}^{\infty} p^{(n)}(x) f(x)\,dx = (-1)^n \int_{-\infty}^{\infty} p(x) f^{(n)}(x)\,dx. \tag{3}$$

Distributions also differ from functions because they are covariant, that is, they push forward. Given a smooth function $\alpha : \Omega_1 \to \Omega_2$, a distribution T on Ω_1 pushes forward to a distribution on Ω_2. In contrast, a real function f on Ω_2 pulls back to a function on Ω_1, namely $f(\alpha(x))$.

Distributions are, by definition, the dual to the smooth functions of compact support, with a particular topology. For example, the delta function δ is the linear functional $\delta(f) = f(0)$. The distribution corresponding to a function g is

$$T_g(f) = \int_\Omega f\, g, \tag{4}$$

and the distribution corresponding to a measure μ is

$$T_\mu(f) = \int_\Omega f\, d\mu. \tag{5}$$

The pushforward map of a distribution T along α is defined by

$$\alpha_* T(f) = T(f \circ \alpha), \tag{6}$$

and the derivative of T is defined by $D\,T(f) = T\left(D^\dagger f\right)$ where $D^\dagger$ is the formal adjoint of D. For example, the first derivative of the delta function is given by

$$\frac{d}{dx}[\delta(f)] = -\left.\frac{df}{dx}\right|_{x=0}. \tag{7}$$

As is the case for any function space, the topology determines which linear functionals are continuous, that is, are in the dual vector space. The topology is defined by the family of seminorms,

$$N_{K,\alpha}(f) = \sup_K \left\| D^\alpha f \right\|, \tag{8}$$

where sup denotes the supremum. It agrees with the C-infty topology on compact subsets. In this topology, a sequence converges, $f_n \to f$, iff there is a compact set K such that all f_n are supported in K and every derivative $D^\alpha f_n$ converges uniformly to $D^\alpha f$ in K. Therefore, the constant function 1 is a distribution, because if $f_n \to f$ then

$$T_1(f_n) = \int_K f_n \to \int_K f = T_1(f). \tag{9}$$

SEE ALSO

Convolution, Delta Function, Delta Sequence, Fourier Series, Functional, Linear Functional, Microlocal Analysis, Statistical Analysis, Tempered Distribution, Ultradistribution

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This entry contributed by Todd Rowland

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